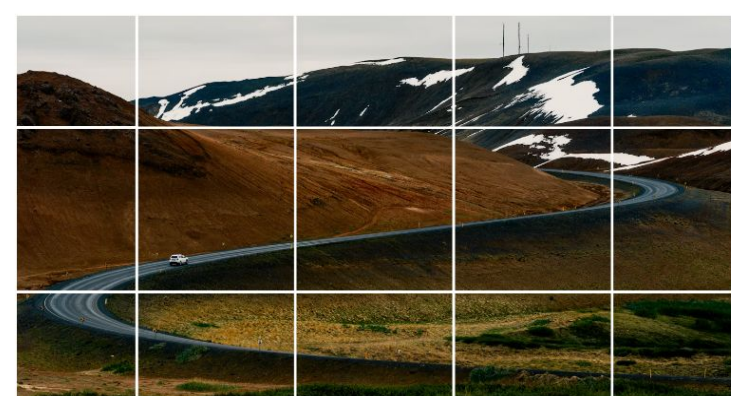
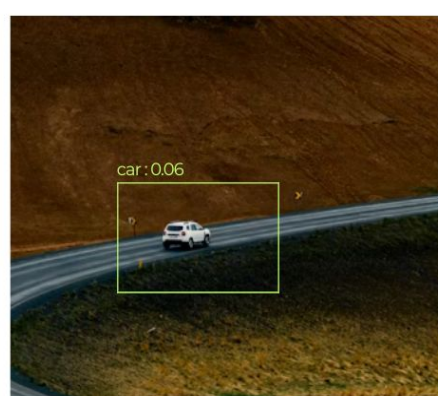


## Abstract

- Autonomous vehicles often struggle to detect small or distant objects, limiting safety.
- Challenge: “How far is too far before an AV loses sight?”
- We present a cloud-assisted pipeline combining stereo depth, tile-based offloading, super-resolution, and YOLO-based perception.
- Extends perception range for reliable detections.
- Selective offloading of tiles conserves bandwidth and latency.



original frame

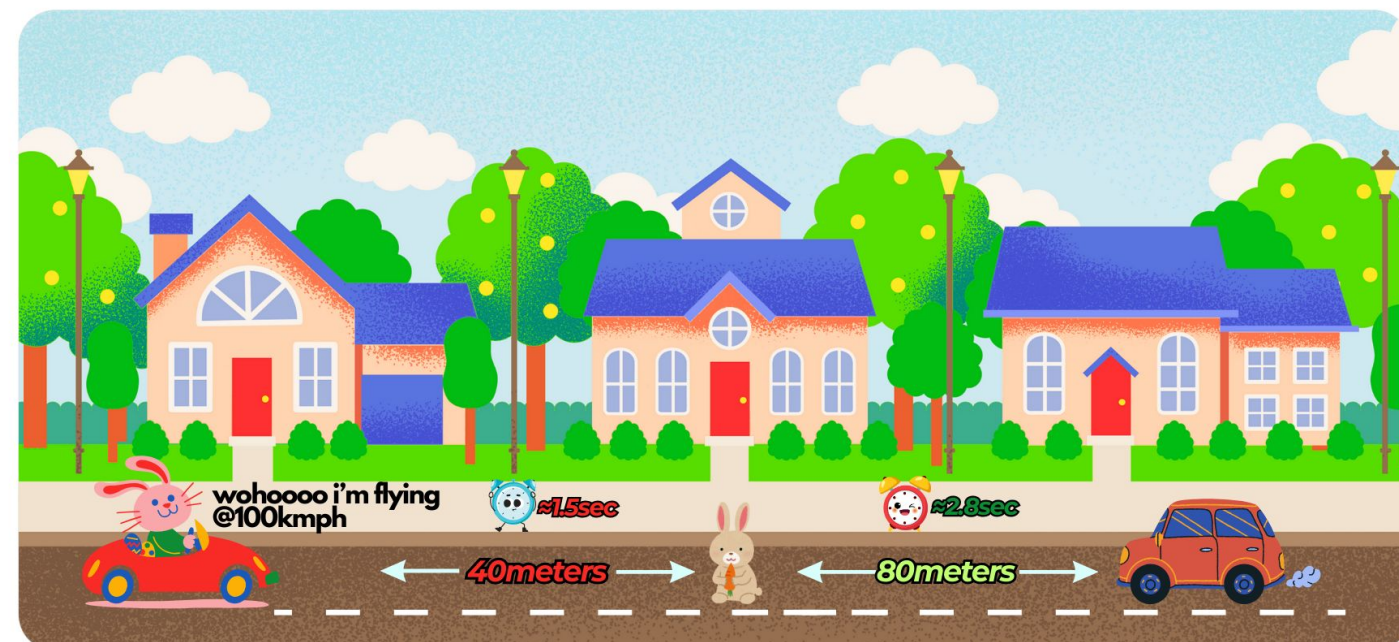


super-resolved tile

- Generalizable Framework that can be extended to tracking and prediction tasks.
- Achieved strong performance:  
Tested on CARLA & KITTI datasets.

## Why To Even Bother?

- AVs must see far to stay safe.
- Distant objects often appear tiny & blurry → YOLO misses them.
- Missing them until they're close = dangerous.
- At highway speeds (100 km/h), the car travels ~28 m every second..



- Safety Buffer:
  - Urban: short distances, frequent stops → okay with near detection.
  - Highways: long braking distances + fast speeds → need early detection → safer maneuvers.

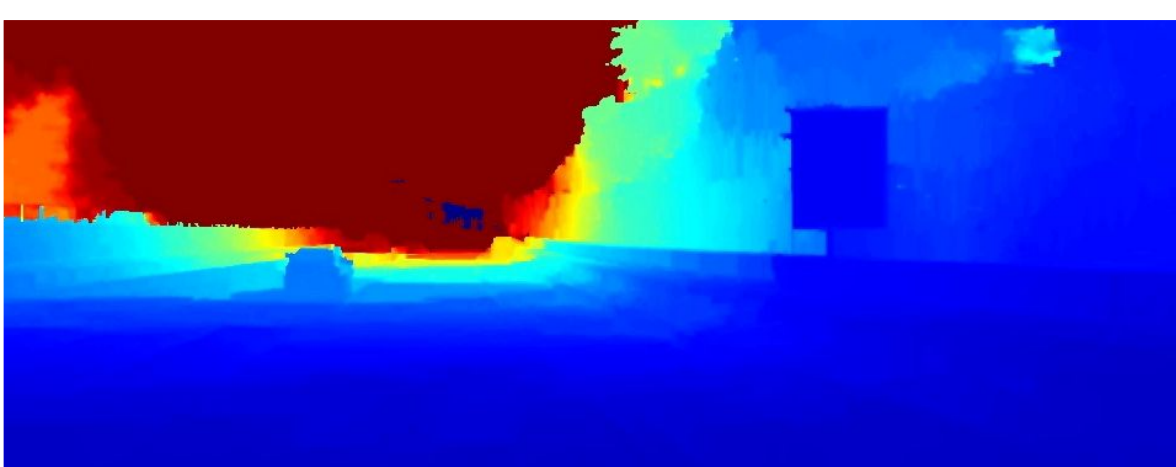
## How This Works?

- Vehicle captures stereo frames.



- Stereo disparity → depth map

$$\text{depth} = (\text{baseline} \times \text{focal\_length}) / \text{disparity}$$



- Distant ROIs identified (objects YOLO misses).

$$\text{far\_mask} = (\text{depth\_map} > \text{distance\_threshold})$$

if depth\_map[pixel] > 40.0: pixel = 1 (far), else: pixel = 0 (near)



- ROIs cropped as Tiles and sent to cloud

Cropped Tile(600x301px)



- Cloud performs super-resolution + detection.

SuperResolved Tile (2400x1504px) → YOLO Inference on SR Tile



- Results mapped back locally.



- Architecture Components:
  - StereoSGBM for depth estimation.
  - YOLO11x for detection.
  - SR model (CARN)

## So Did It See Far Enough?

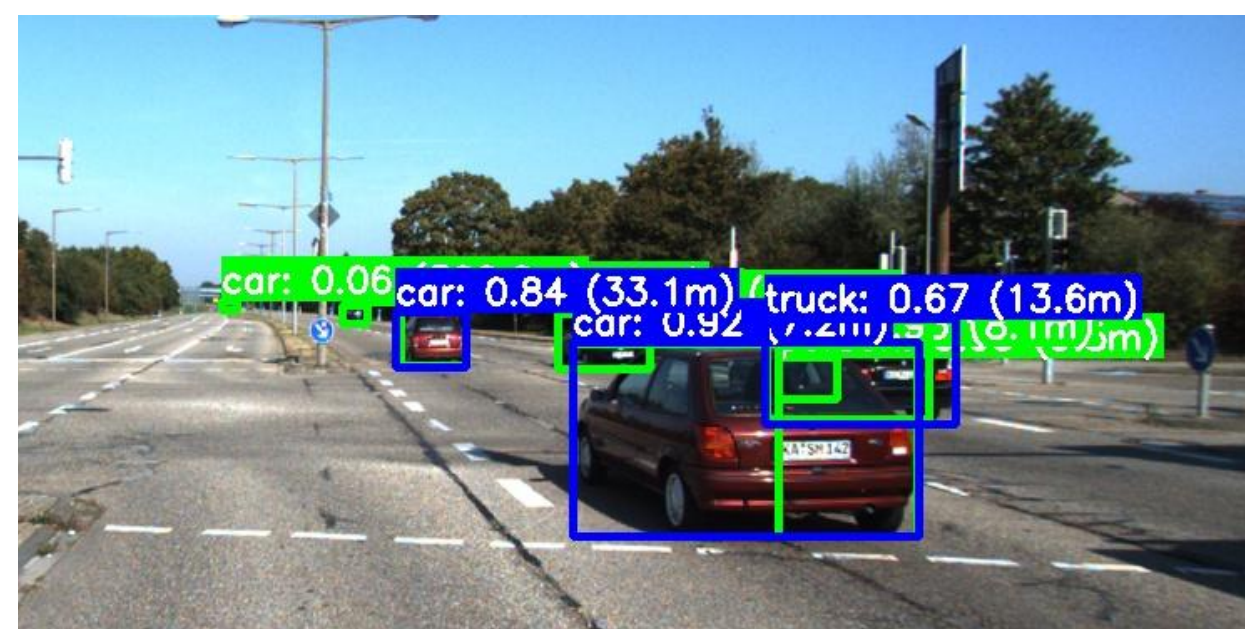
- Without Stereo-SuperRes (Basic Yolo Detection)

Baseline YOLO fails to detect distant vehicles  
[○ : Not detected]

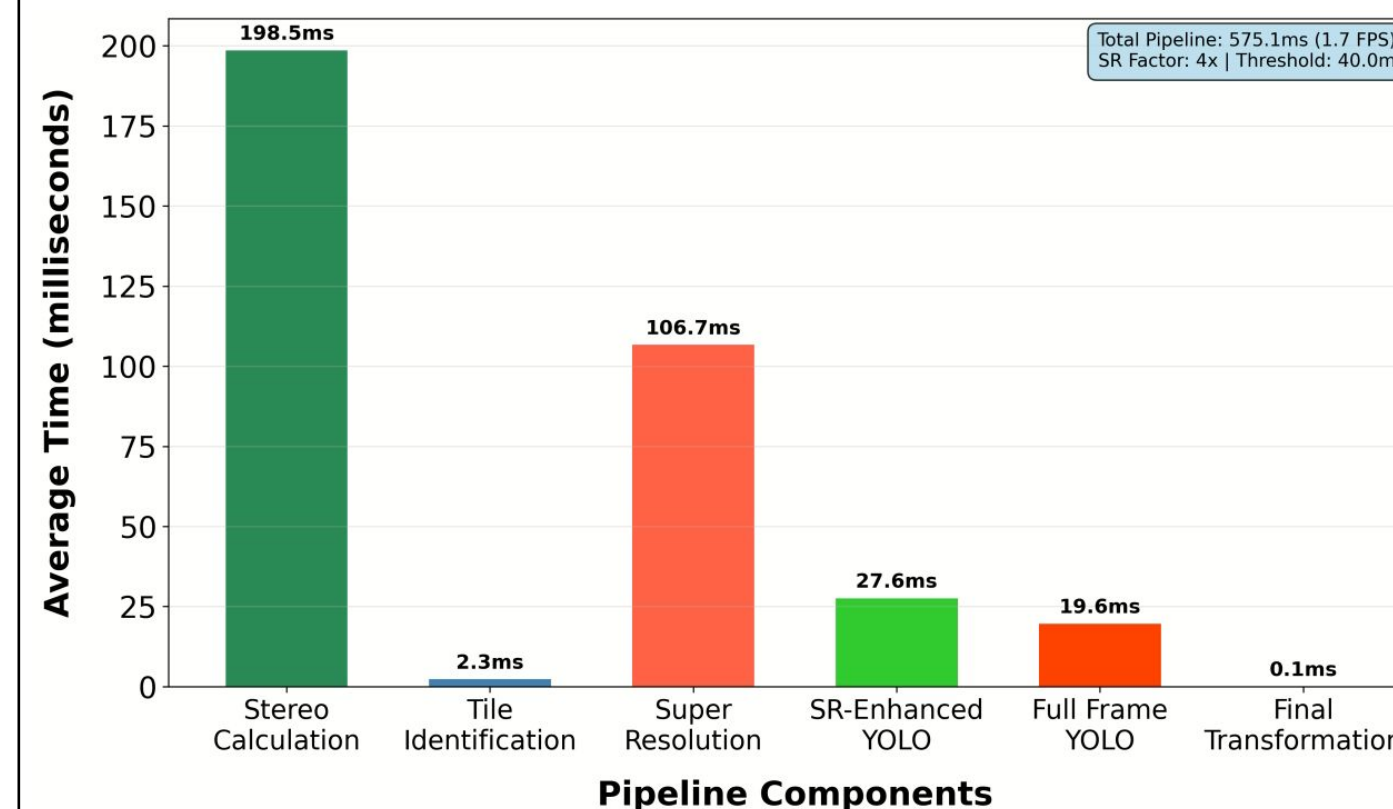


- With Stereo-SuperRes

Stereo + Super-Resolution extends detection range  
[□ : Detected using SuperRes]



- Pipeline Timing Analysis:



- Pipeline is ≈ 28% slower than baseline YOLO
- Detection improved by 136.9%

- Stereo Accuracy:

- KITTI dataset: MAE ≈ 5.07m

## Challenges In The Approach

- Tile selection accuracy: If stereo depth misidentifies ROIs, some small objects may still be missed.
- Super-resolution artifacts: SR may sometimes hallucinate details → risk of false detections.

## Conclusion

- Proposed a cloud-assisted pipeline that extends perception range in autonomous vehicles.
- Combined stereo depth + selective tile offloading + super-resolution + YOLO detection.
- Achieved more reliable small/distant-object detection compared to baseline methods.
- Demonstrated feasibility on CARLA & KITTI datasets.
- Shows promise for safer highway driving and resource-efficient cloud integration.
- Provides a generalizable framework that can be extended to other vision tasks (night vision, fog, low-res cameras).

## Future Direction

- Adapt the pipeline to work on videos in real time while balancing detection accuracy and computational efficiency.
- Comparisons to explore:
  - Stereo vs Monocular depth.
  - Impact of different SuperResolution models.
- Pipeline Enhancements:
  - Confidence-based selective SR offloading.
  - Bandwidth/latency analysis of ROI-based cloud streaming.
  - Dynamic tile sizing: smaller crops for small objects, larger for groups.
  - Tile Tracking Over Time: Instead of re-selecting far ROI every frame, track the same tiles across consecutive frames for efficiency.
- Extending Capabilities:
  - Stress-testing in adverse weather (rain, fog, low light) to check robustness.

## References

- Geiger, A., Lenz, P., & Urtasun, R. KITTI Vision Benchmark Suite.
- Ahn, N., Kang, B., & Sohn, K.-A. CARN: Cascading Residual Network for Fast, Accurate Image Super-Resolution. ECCV 2018.

## Have a Chai with the Authors



Ritik Vaishnav  
Research Intern



Najiya Naj  
PhD Scholar



Dr. Arani Bhattacharya  
Assistant Professor